



Investigation of lithium-ion battery degradation mechanisms by combining differential voltage analysis and alternating current impedance

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HIGHLIGHTS

- 18650-type Cells are cycled at 0 °C and 25 °C using two charging protocols.
- Main degradation factors are loss of lithium inventory (LLI) and active cathode loss.
- Neutron powder diffraction and post-mortem analysis are done for deep understanding.
- Correlations between material loss and impedance parameters are revealed.
- Warburg impedance coefficient could be correlated to LLI in the course of cycling.

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ABSTRACT

18650-type cells with 2.5 Ah capacity are cycled at both 25 °C and 0 °C separately, and at 25 °C two charging protocols (constant current, and constant current-constant voltage charge) are used. Differential voltage analysis (dV/dQ) and alternating current (AC) impedance are mainly used to investigate battery degradation mechanisms quantitatively. The dV/dQ suggests that active cathode loss and loss of lithium inventory (LLI) are the dominating degradation factors. Significant microcracks are observed in the fatigued cathode particles from the scanning electron microscopy (SEM) images. Crystal structure parameters of selected fatigued batteries at fully charged state are determined by *in situ* high-resolution neutron powder diffraction. Obvious increases of ohmic resistance and solid electrolyte interphase (SEI) resistance occur when the battery capacity fade falls beneath 20%. Continuous charge transfer resistance and Warburg impedance coefficient (W_{eff}) increase are observed in the course of cycling. Correlation analysis is performed to bridge the gap between material loss as well as LLI and impedance increase. The increase of the charge transfer resistance is related to both active cathode loss and LLI, and a functional relationship is revealed between LLI and W_{eff} regardless of the used cycling protocols.

1. Introduction

The lithium-ion batteries are nowadays the technology of choice for replacing the fossil “energy storage” in modern transportations, e.g. electric vehicles (EVs) and hybrid electric vehicles (HEVs), due to their outstanding energy and power density [1–4]. For vehicular applications, battery internal resistance rise and capacity fade over calendar and cycle life are still the major barriers [5–8]. Understanding the relevant

degradation mechanisms in batteries is of great importance to optimize the battery lifetime [9–14]. Several degradation mechanisms, namely loss of cyclable lithium due to the growth of solid electrolyte interphase (SEI) [15] at the anode caused by electrolyte decomposition and lithium consumption [16–20], the loss of active material caused by mechanical stress due to structural changes of cathode and anode [21–27], impedance increase [3,6,28–30] and their combinations are used to interpret the deterioration of battery performance.

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To reveal the contributions of different degradation mechanisms for commercial cells, two quantitative analytical methods are commonly employed: the differential voltage analysis (dV/dQ) and the alternating current (AC) impedance analysis. Both methods have the advantage that the average information of the degradation can be identified without opening the cells. The dV/dQ method, which is obtained from the differentiation of the cell charge/discharge voltage with respect to state of charge (SoC), has been widely reported for $\text{Li}(\text{NiCoMn})\text{O}_2$ (NCM)/graphite [31,32], $\text{Li}(\text{NiCoAl})\text{O}_2$ (NCA)/graphite [33,34], NCM blended with LiMn_2O_4 (LMO)/graphite [16,18], LiCoO_2 (LCO)/graphite [35,36], and LiFePO_4 (LFP)/graphite batteries [14,37–39]. The peaks on the dV/dQ curves usually represent material phase transitions [16], the peak shift and the changes in the peak to peak capacity are used to estimate the contributions of different fatigue mechanisms, e.g. loss of lithium inventory (LLI), active cathode loss, and anode loss [31–33,35–38]. The AC impedance provides information on the cell degradation processes occurring in the frequency domain (at various time scales) [40]. Battery internal dynamics, e.g. the growth of SEI, charge transfer reaction, lithium ion diffusion, can be identified with properly designed impedance models [12,29,41]. By comparing the impedance of the pristine and degraded cells, the degradation mechanisms can be quantified because some identified kinetic parameters usually act in cooperation with the battery capacity fade [6,11,28].

For the commercial battery degradation study, researchers are either using the dV/dQ due to its ability of separating the cathode and anode curves by their unique features, or using AC impedance on account of its faster measurement and its potential of the online implementability [42–44]. Studies reporting the combination of the above-mentioned two methods are rather rare. Within the dV/dQ framework there is no obvious way to seek out the battery power fade by resistance rise, *vice versa*, some degradation modes, e.g. active material loss, mobile lithium loss, are hard to be identified only using impedance. Schindler and Danzer [39] introduced a framework combining the electrode balancing and degradation pathways based on the correlation between altered impedance features and available lithium loss as well as active material loss, and the coupling between thermodynamic and kinetic degradation modes is proposed for the LFP/graphite battery. The thermodynamic degradation modes are defined as the reduction of capacity under equilibrium conditions, which can be described by the open circuit voltage (OCV) curve. The kinetic degradation, which is usually identified by the impedance, is expressed by an increase of overpotential added to the OCV curve depending on the applied current [39]. The change of charging and discharging voltage curve contains both thermodynamic and kinetic characteristics of the battery. Recently, Severson et al. [14] reported a promising route that machine learning can be used to construct models that accurately predict battery lives, using voltage data collected from charge-discharge cycles of the LFP/graphite battery. The $\text{Li}_x(\text{TM})\text{O}_2$ (TM, transition metal = Ni, Co, Mn, Al) cathode usually exhibits different degradation mechanisms compared to other cathodes. The correlations between thermodynamic degradation and kinetic degradation factors in the $\text{Li}_x(\text{TM})\text{O}_2$ batteries are still under discussion [34]. Bridging this gap will provide deeper understanding of degradation mechanisms in commercial lithium-ion batteries, which is essential to improve the control strategies in battery management systems to prolong the battery service time in automobiles.

Two primary objectives are encompassed in this work, the first goal is to quantify the degradation mechanisms by using dV/dQ and AC impedance for 18650-type cells which are cycled at 25 °C and 0 °C, respectively. The battery is comprised of 42 (3) wt.% NCM - 58 (3) wt.% NCA (see Fig. S1) as cathode and graphite as anode. *Post-mortem* analysis using scanning electron microscopy (SEM) and *in situ* neutron powder diffraction for selected fatigued cells are employed to assist the battery degradation mechanism analysis. The second purpose is to bridge the gap between thermodynamic degradation factors from dV/dQ and kinetic degradation factors from impedance spectroscopy and to find their correlations. It will accelerate the identification of the relevant

degradation mechanisms for battery diagnostic and prognostic applications.

The paper is organized as follows. Firstly, the experimental tests are described in Section 2. Section 3.1 presents the capacity fade results of the cycled cells. The methodology of the dV/dQ method followed by degradation mechanism analysis is introduced in Section 3.2. The *post-mortem* analysis using SEM is performed in Section 3.3. The application of *in situ* neutron powder diffraction is presented in Section 3.4. The impedance spectroscopy results are shown in Section 3.5, and the mathematical correlation analysis is depicted in Section 3.6. Section 3.7 summarizes the degradation mechanisms during the capacity fade and Section 4 provides conclusions.

2. Experiments

2.1. Cycling procedures of 18650-type commercial cells

The 18650-type cells with nominal capacity of 2.5 Ah are cycled, and at least two cells for each fatigue stage are used to check the reproducibility. The specifications of the battery are elucidated in Table 1. The constant current (CC) and the constant current-constant voltage (CC-CV) charging techniques at different temperatures are usually used in the literature [45,46]. In the present paper, the cells are treated differently and can be divided into three groups:

- a: Cycling with CC protocol at 25 °C (CY25-CC)
- b: Cycling with CC-CV protocol at 25 °C (CY25-CC-CV)
- c: Cycling with CC protocol at 0 °C (CY0-CC)

The cycling condition of the CC protocol is a 2.5 A (1C) current for the charge with the 4.2 V voltage cut-off. In case of CC-CV protocol, the cells are charged at a constant current of 2.5 A until a cell voltage of 4.2 V is reached, followed by a constant voltage period until the current drops below 0.1 A (C/25). Discharge is carried out with constant current 2.5 A until cell voltages of 2.5 V are reached in all cases. 10 min rest is set between charging and discharging during the cycling. The batteries are cycled up to 700 times at 25 °C and up to 1100 times at 0 °C.

2.2. Calibration experiments

The cycling tests are interrupted in 100 cycle intervals for calibration in order to obtain the residual capacity, differential voltage profile as well as the cell impedance. The scheme of the calibration protocol is presented in Fig. S2. To get the residual capacity, the battery is fully charged with C/2 up to 4.2 V, followed by constant voltage (CV) charging to a cutoff current of 0.1 A. After 1 h relaxation, the cell is discharged at 1C with the discharge cutoff voltage 2.5 V suggested by the manufacturer. The discharge capacity using 1C current serves as the residual capacity in our research to evaluate the battery degradation. After being fully charged again, a low current discharge at C/25 is carried out until a cutoff voltage of 2.5 V is reached, which is used for the dV/dQ in our study. The impedance spectroscopy characterization is performed at approximately 80% SoC, 50% SoC, and 20% SoC with a current amplitude 1 A in the frequency range between 10 kHz and 0.01 Hz at 25 °C. The SoCs are determined by the discharging process using 1

Table 1
Specifications of the 18650-type lithium-ion battery.

Battery type	18650
Anode material	Graphite
Cathode material	Blend of 42 (3) wt.% NCM and 58 (3) wt.% NCA (Fig. S1)
Electrolyte	Solution of lithium hexafluorophosphate (LiPF_6)
Nominal voltage	3.6 V
Cutoff voltage	2.5 V ~ 4.2 V
Nominal capacity	2.5 Ah
Battery mass	45.0 g

C current and referring to the residual capacity of each cell after being fully charged. For example, for the fatigued cell with 2.0 Ah residual capacity, the 50% SoC is reached by discharging for 24 min using 2.5 A after being fully charged. To achieve the electrochemical equilibrium of the battery, 1 h rest is set before the impedance test. The Chroma 17020, BioLogic BCS and VMP testers are employed for cell cycling and calibration tests as well as impedance tests. All measurements are conducted at 25 °C in a climate chamber.

2.3. Cell opening and electrochemical investigation

The cell at the beginning of life (BoL) and three fatigued cells are disassembled at fully discharged state under argon atmosphere in a MBraun glove box. The disassembled cells are CY25-CC-700 (700 cycles), CY0-CC-400 (400 cycles), and CY0-CC-1100 (1100 cycles). The morphology of the cathode and anode is obtained from scanning electron microscopy (SEM) from Zeiss Merlin. For all electrochemical experiments, one side of the electrode coating is removed using N-methyl-2-pyrrolidone (NMP) and the residual electrode is washed in dimethyl carbonate (DMC) and punched out in discs with a 12 mm diameter. For each case two CR2032 type coin cells are assembled for independent electrochemical characterization as reproducibility tests. The half-cells have the anode or cathode as working electrode and metallic lithium as a counter electrode. The 3-electrode cells include the cathode and anode plus metallic lithium as reference electrode to monitor the potential change of the cathode vs. Li^+/Li and anode vs. Li^+/Li . The Celgard separator is used for the coin cells. 1 M lithium hexafluorophosphate (LiPF_6) in a mixture of dimethyl carbonate (DMC) and ethylene carbonate (EC) at a volume ratio of 1:1 is used as electrolyte. In the later part of this paper, the terms “half-cell” and “3-electrode cell” are employed to express the two type of assembled cells. The 3-electrode measurements are performed using the same cutoff voltage as for the commercial cells, i.e. between 2.5 V and 4.2 V. The cathode and anode half-cell experiments are conducted in potential ranges of 4.25 V–2.7 V vs. Li^+/Li and 2 V–0.01 V vs. Li^+/Li , respectively. Slow-scan cyclic voltammetry is carried out at cathode half-cell at a scanning rate of 0.025 mVs^{-1} . The voltage and differential voltage curves of cathode, anode and 3-electrode cell are obtained by cycling the assembled cells with 0.1 mA current at 25 °C. The impedance is tested with sinusoidal voltage signal 5 mV for the half-cells at different SoCs. The excitation frequency is set ranging from 1 MHz to 0.001 Hz to get more impedance information.

2.4. In situ neutron powder diffraction

Collection of the neutron powder diffraction data is carried out at fully charged state for the BoL cell, CY25-CC-400 (400 cycles), and CY25-CC-700 (700 cycles), respectively. High-resolution neutron powder diffraction data are collected at diffractometer SPODI [47]. Measurements are performed at ambient temperature using monochromatic neutrons with $\lambda = 1.5481 \text{ \AA}$ and $\lambda = 2.5360 \text{ \AA}$ obtained from a vertically focused composite Ge monochromator in (551) and (331) orientation. Operando neutron powder diffraction experiments are planned for the future at finely controlled temperatures within the framework of the “Energy research with Neutrons (ErWiN)” project [48]. All cells are firstly charged with 0.4 A up to 4.2 V, followed by constant voltage charging down to a cutoff current of 0.04 A (C/62.5). Data is recorded after reaching the desired charge state and an additional time of relaxation. The Rietveld method is employed for the structure refinement of the different phases consisting in the diffraction pattern, and the instrument resolution function is determined using a $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$ standard. The Fullprof software is used for the Rietveld refinement [49]. Additional information about the neutron powder diffraction test is summarized in the supplement.

3. Results and discussion

3.1. Capacity degradation (capacity fade)

The residual capacity of the cells after the cycling tests is shown in the Fig. 1a as a function of cycle number. Error bars represent standard deviations obtained by the deviations between at least two cells cycled at identical conditions. In Fig. 1b, the capacity loss is plotted as a function of total capacity throughput. Capacity throughput represents the amount of charge delivered by the battery during cycling. The charge and discharge voltage profiles for getting the residual capacity are presented in Fig. 1c, in which 700 and 1100 are the cycle numbers. Users are also concerned with the energy which can be extracted from the cells. Fig. 1d displays the battery energy (integral of voltage multiplied by charge) as a function of cycle numbers. For the three different cycling conditions, it can be observed that the energy curves differ after 400 cycles.

Apparently, when compared at the same cycle number, it can be concluded that the capacity decreases linearly within the first 400 cycles for all three groups, and the CY25-CC-CV cells show a faster capacity degradation rate than the CC protocol cells in Fig. 1a. It drops to 65% of nominal capacity for CY25-CC-CV and to 75% of nominal capacity for CY25-CC after approximately 700 cycles. The capacity fade of CY0-CC cells shows a slower degradation rate between the 400th to the 1000th cycle, and further cycling to 1100 cycle numbers leads to a faster degradation. At the end of the cycling test (1100 cycles), 15% capacity loss is observed. Usually, side reactions like underlying lithium plating happen at subzero temperature [50,51], meaning the low temperature has a faster capacity loss than the room temperature. But, in our research, by comparing the capacity vs. cycle plot in Fig. 1a, the CY0-CC has a similar capacity loss with the CY25-CC until the 400th cycle, and the CY0-CC shows a better performance after 400 cycles. By comparing the capacity vs. capacity throughput in Fig. 1b, the CY0-CC degrades slightly faster than that of the CY25-CC before 1500 Ah, and the degradation rate of CY0-CC slows down after the 1500 Ah. Two possible explanations are given briefly as follows. The first one is that the cells are self-heated during cycling. The voltage curves and the corresponding temperature profiles of the first three cycles are shown in Fig. S3a. It indicates that the surface temperatures of CY25-CC and CY25-CC-CV are similar, their temperatures increase to 27 °C in the charging process and to 30 °C during discharging. The maximum temperature of the CY0-CC cell is 6 °C in the charging process and is 9 °C at the end of discharging. Thus, it is hypothesized that the self-heating behavior allows the CY0-CC cell escaping from the serious side reactions. The second reason causing the capacity deviation might be the different usage of battery charge and discharge depth. The voltage vs. capacity curves for all three groups at the beginning of the cycling are compared in Fig. S3b. It can be seen that the cells cycled at 0 °C have a narrower working range compared to the cells cycled at 25 °C. The BoL impedance curves at different temperatures are compared in Fig. S3c. As the temperature decreases, dramatic increase of the impedance arc can be observed, which indicates that the polarization takes effect on the terminal voltage during the cell cycling. It causes that the actual operation depths of charge/discharge are essentially different for the three cycling protocols. As a general trend, the capacity of the cells degrades faster when cycled at higher voltage level, and the degradation rate of the cells cycled at narrower depth of discharge (DoD) is substantially lower than that at wider DoD [10,31]. It might be the reason that the CY0-CC has less capacity loss after 400 cycles (or 1500 Ah) shown in Fig. 1a (or Fig. 1b). In summary, the difference of capacity loss for the three cycling protocols, as illustrated in Fig. 1, results from the effect of temperature and depths of charge/discharge comprehensively. The capacity loss mechanisms are further examined by applying dV/dQ and AC impedance methods.

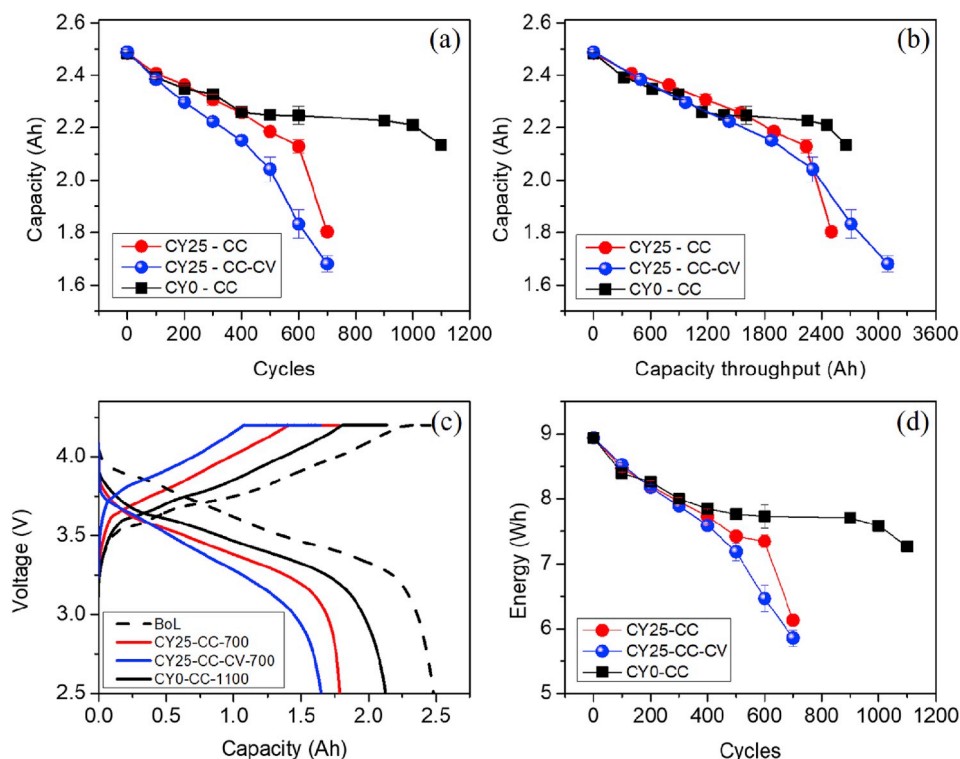


Fig. 1. Selected electrochemical data for 18650 cells cycling under three different conditions. (a) Residual capacity of cells as a function of cycle number, (b) and as a function of total capacity throughput. (c) Voltage profiles for different cells. (d) Energy of cells as a function of cycle number.

3.2. Degradation mechanisms by dV/dQ method

The most straightforward information provided by battery charging and discharging, usually determined using rate of 1 C, is the residual capacity. In order to extract as much information as possible, the dV/dQ method, obtained simply by differentiating the voltage with respect to the capacity at low current, is introduced. The thermodynamic degradation factors (active cathode loss, active anode loss, and LLI) of the electrode, namely the reduction of its capacity under near equilibrium state, can be obtained from the dV/dQ curve.

3.2.1. dV/dQ methodology derived from coin cell tests

Fig. 2a and Fig. 2b show the cathode vs. Li^+/Li , anode vs. Li^+/Li , and full cell (cathode vs. anode) curves, and their corresponding derivatives ($|dV/dQ|$ curves), respectively. The voltage positions turn to the obvious peaks by differentiating the voltage with respect to the capacity, and their corresponding positions are marked in Fig. 2 and summarized in Table 2.

The dV/dQ curve of the anode exhibits three characteristic peaks, marked as AN1, AN2 and AN3. Kato et al. [17] have performed a detailed investigation on the lithiation/delithiation process in the graphite anode by *in situ* XRD measurements. They found that the AN1 peak was observed at $x \approx 0.58$ in Li_xC_6 derived from the insufficient supply of lithium in LiC_6 inside the graphite particles. A single phase of LiC_{12} at the anode was observed when the lithium content reached $x \approx 0.2$ around the AN2 position. In contrast to the characteristic dV/dQ profile of the anode exhibiting several narrow peaks and zero values beyond the peaks, the absolute value of the cathode dV/dQ is always larger than zero, resulting from the gradual decrease of cathode potential during the cell discharge. Two broad peaks from the cathode, which are related to the electrochemical activity of the cathode, are observed in Fig. 2b and marked as CA1 and CA2. The cathode of the cell consists of two phases of NCM and NCA, which have similar crystal structures (layered oxides) and exhibit electrochemical activity at very similar potentials as reported in Ref. [22]. The main activity windows of the

layered oxide NCA and NCM are mostly overlapping in the range of 3.5 V–4.2 V, and the incremental capacity curve of a blended cathode is the sum of each single phase incremental capacity [22,23]. The cyclic voltammetry curve of the blended cathode is shown in Fig. 2c. It can be observed that the CA1 and CA2 are all located in the main electrochemical activity region, indicating that the broad CA1 and CA2 at 4.081 V and 3.897 V, respectively, are attributed to the combination of the lithium ion intercalation into the NCA and NCM. And the potential corresponding to the CA1 peak is approximately located at the potential where the ratio of c/a lattice parameters of NCM and NCA have reached a maximum, and at this potential, the partial oxygen reduction of NCM and NCA occurs [22]. A sole leftward shift of the anode voltage curve, which can be illustrated by the L0 increase as marked in Fig. 2a, is mainly caused by the reduction of the inventory of cyclable lithium and change of the working window of the electrodes [33]. The important hypothesis for using the dV/dQ method is that the distance between any of these peaks is directly proportional to the active material capacity [16]. L1 is the distance between the first point of discharge and the peak CA1. The cathode loss is estimated by the reduction of L1 in the fatigued cell, relative to that of the BoL. The characteristic peaks of the anode (AN1, AN2) correlate to the phase transformations of lithiated graphite Li_xC_6 during cycling. The distance L2 between the anode peaks is used to obtain the information about the storage capabilities of the anode material. Because only the terminal voltage can be controlled in the battery, it is hard to locate the end points of cathode and anode in the full cell (dashed curves in Fig. 2a), the leftward shift of anode peak AN2, which is used in Refs. [16,33], is employed to describe the operating window alteration of the electrodes and quantify the total cyclable lithium. Consequently, the LLI is expressed by the change of L3.

The dV/dQ method provides the description of the cell state without performing an invasive investigation, but there are some limitations in quantifying the electrode degradation. The assumption that the degradation between any of the peaks is directly proportional to the active material capacity neglects the possibility of uneven degradation. Additionally, the degradation rates of NCM and NCA are hard to distinguish

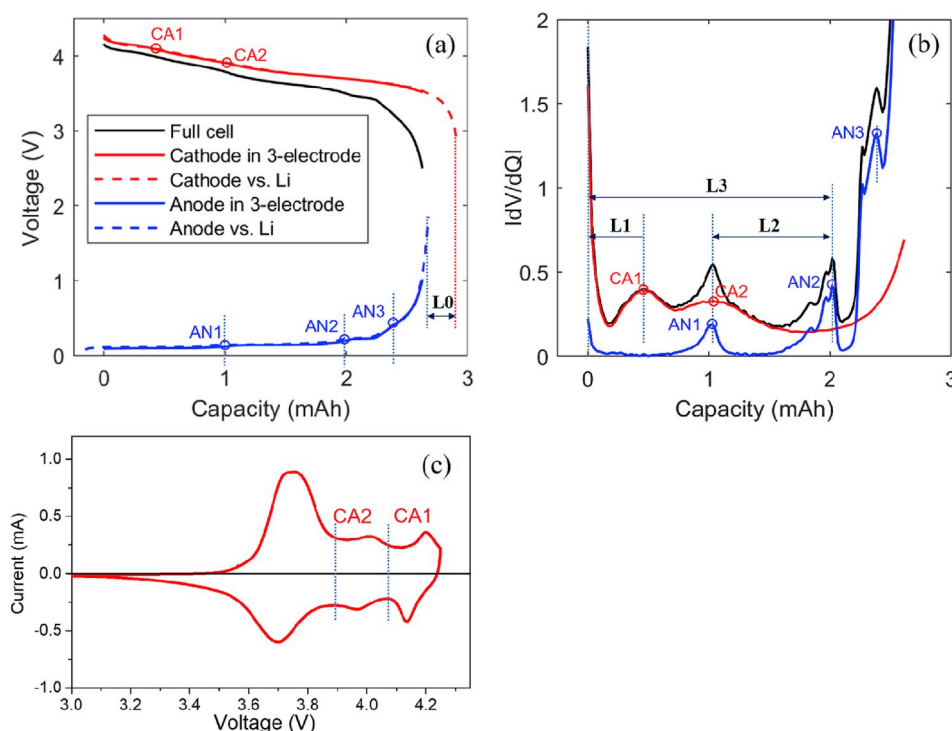


Fig. 2. Electrochemical information of the half-cell and 3-electrode cell. (a) Voltage vs. capacity curves. (b) dV/dQ vs. capacity curves during discharge measured in three-electrode setup. (c) The cyclic voltammetry curve of the blended (NCM and NCA) cathode. Dashed curves in (a) are the cathode and anode curves measured in half-cell measurements, which are aligned to the corresponding curves measured with 3-electrode cell measurements. CA1 and CA2 are characteristic features of the cathode, AN1, AN2 and AN3 are characteristic features of the anode.

using this method.

3.2.2. 18650-Type cell results

The voltage curves using C/25 discharge current are plotted in Fig. S4, and their corresponding $|dV/dQ|$ curves for one selected cell in each fatigue stage are well defined. The percentage of cathode loss, anode loss and LLI are summarized in Fig. 3.

As presented in Fig. 3, the change trends of LLI (Fig. 3a) and cathode loss (Fig. 3b) all present continuous downward trend. It reveals that anode material loss happens in the first 200 cycles in Fig. 3c, then it remains unchanged for all cells. Its influence is relatively small compared to the other two degradation factors. Until 400 cycles, the LLI of cells cycled at 25 °C and 0 °C exhibits a similar degradation rate, however, the rate of LLI for cells cycled at 25 °C increases faster than the cells cycled at 0 °C after 400 cycles. The drop of LLI after 600 cycles for CY25-CC and after 500 cycles for CY25-CC-CV, as marked by the green arrows, is observed, which is consistent with a sudden increase of the SEI resistance (discussed in Section 3.5). The cathode loss in Fig. 3b exhibits a similar trend as the LLI with the increasing cycle number. From the trend lines, the CY0-CC behaves better than CY25-CC and CY25-CC-CV. And for CY0-CC, both LLI and cathode loss slow down between the 600th and the 900th cycle. Also, for the CY25-CC and CY25-CC-CV, the LLI and cathode loss depict the same degradation trend. The results indicate that the LLI and cathode loss might happen simultaneously. One possible interpretation is that the loss of lithiated lithium sites is the main reason leading to the active cathode loss in our cycling cases, i.e. the lithium is trapped in the cathode during cycling, it leads to both, loss of mobile lithium and loss of active cathode. Another alternative explanation for the simultaneous loss of cathode and lithium is that the cathode particles are fractured after the battery fatigue. The formation of new interface in the cathode will also lead to the irreversible loss of mobile lithium. The LLI and cathode loss are deemed to contribute to the total capacity loss together, and their different contributions behind the cycling protocols lead to the difference of capacity loss in Fig. 1. In later parts, the *post-mortem* and *in situ* neutron powder diffraction methods are used to investigate the underlying degradation mechanisms.

3.3. Degradation mechanisms by SEM

The SEM morphologies of the cathode and anode material from the BoL cell and three typical fatigued cells, i.e. CY25-CC-700, CY0-CC-400, and CY0-CC-1100, are compared in Fig. 4. The positions of NCM and NCA particles are addressed by employing Energy-dispersive X-ray spectroscopy (EDX), and both phases show a dense agglomerated structure where all particles are connected to each other as displayed in Fig. 4a. The size of the secondary particles is 5 μm –10 μm , while the size of the primary particle of all phases is in the range of 250 nm–900 nm. Some microcracks are observed inside the secondary grains of fatigued cathodes, as seen from the Fig. 4c, e, and Fig. 4g. Many particles are too fragile that they become pulverized in the SEM sample as displayed in Fig. 4g. As reported in Refs. [52–55], high anisotropic strain generated with repeated lithium insertion and extraction of the host lattice are believed to be responsible for the nucleation of microcracks that eventually causes the disintegration of the oxide particles. The microcracks in the cathode particles open new channels for the electrolyte and this expediting its infiltration into the particle interior, which could lead to continuous accumulation of NiO-like impurity layer on the new internal surfaces [53]. The microcracks can also cause transport barriers for electrons and lithium ions, and even lead to contact loss with active material inactivity partially. The process of the loss of electronic contact among the active material also increases the cell internal impedance [53,56]. Additionally, the microcracks in the secondary particles contribute to capacity loss by causing the additional cathode loss and mobile lithium loss due to the electrolyte penetration into the particle leading to the formation of new cathode electrolyte interface [22,55]. Thus, the observations from SEM support the obvious cathode loss in dV/dQ results in Fig. 3b. The fatigued anode reveals less pronounced structure changes from the surface morphology in Fig. 4. It confirms that the anode performance does not change significantly during fatigue. Few fatigued anode particles burst on the surface of the particles as marked in the dashed circles, which is assumed to be in accordance with the anode material loss in the first 200 cycles.

Table 2Positions of dV/dQ curve peaks of cathode vs. Li^+/Li and anode vs. Li^+/Li .

	Capacity (mAh)	Voltage (V)	$ dV/dQ $ (V mAh ⁻¹)
CA1	0.449	4.081	0.397
CA2	1.033	3.897	0.359
AN1	1.016	0.122	0.186
AN2	2.034	0.214	0.393
AN3	2.402	0.418	1.282

3.4. Degradation mechanisms investigated by neutron powder diffraction

The neutron powder diffraction pattern for the BoL cell including the Rietveld refinement is shown in Fig. 5a. Vertical bars placed in rows (1)–(7) mark Bragg peak positions of the refined phases. (1) – cathode NCM; (2) – cathode NCA; (3) – copper current collector; (4) – steel housing; (5) – aluminum current collector; (6, 7) – lithium intercalated carbons LiC_{12} and LiC_6 respectively.

No new phases appear and no phases change for the cathode after the cell cycling, which is consistent with the previous reports [22,57]. The 003 peaks of NCA and NCM are enlarged as shown in Fig. 5b. It can be seen that the relative intensity of NCM reduces after the cell degradation, and slight structural changes can be observed from the peak shift. The refined lattice parameters are listed in Table 3, and the unit cell volumes are plotted in Fig. 5d. An increasing trend of the unit cell volume can be observed in the course of cycling guided by the dashed lines. As illustrated in our previous report [22,27,55], which is also consistent with Ref. [58], the lattice parameters in the layered transition metal oxides are highly depended on the lithium content. The unit cell volume decreases monotonously during the battery charging (lithium extraction from the cathode), and a drastic volume decrease can be observed beyond 4.1 V (NCM vs. Li^+/Li) because of the reduction of the repulsion between the oxygen layers [22]. As described in Section 2.4, all selected fatigued cells for neutron powder diffraction test are fully charged to 4.2

V followed by constant voltage charging down to a cutoff current of 0.04 A (C/62.5), meaning that all cyclable lithium in the fatigued battery has been extracted from the cathode. The observed increase of unit cell volume in Fig. 5d means that some increasing amount of lithium is still intercalated in the NCM/NCA structure, resulting in a larger unit cell volume. This result is in good agreement with the loss of lithium inside the cathode structure. The trapped lithium in the cathode will not be used again under the controlled battery voltage window (2.5 V ~ 4.2 V). This lithiated part of the cathode becomes inactive, which is deemed to cause the cathode material loss and lithium loss at the same time.

It is known that in the fully charged state of a commercial battery, the formation of the lithium intercalated carbons LiC_6 and LiC_{12} can be observed. Rietveld refinement of the lattice parameters of these phases does not show any significant changes with battery fatigue anode as shown in Table S1, indicating the structural stability of these stages. Comparing the 001 peak intensity of LiC_6 and 002 peak of LiC_{12} for BoL and fatigue cells in Fig. 5c, it can be found that the weight fraction of the LiC_6 phase decreases, whilst the amount of LiC_{12} phase increases. The small 002 graphite peak occurs in Fig. 5c, corresponding to not fully lithiated anode in the fatigued cell. From Fig. 5e, the quantitative evaluation of the BoL reveals a weight ratio of LiC_6 to LiC_{12} as approximately 11.0:1. Upon cycling this ratio changes to 2.7:1 and to 1.1:1 for the CY25-CC-400 and CY25-CC-700 cells, respectively. These phase fraction changes correspond to a reduction of the amount of active lithium inserted into the graphite, which has been proven in the Ref. [26,59]. The lithium content x in Li_xC_6 is calculated by the molar percentage as presented by Equation (1) ~ (3). The quantitative recalculation of the lithium amount x inside the anode at the fully charged state presents a reduction of 10% and 22% for CY25-CC-400 and CY25-CC-700 cells respectively, which show good agreement with the respective LLI estimated by dV/dQ .

$$\text{mol}\%(\text{LiC}_6) = (\text{Wt}\%(\text{LiC}_6) / 79.0052) / (\text{Wt}\%(\text{LiC}_6) / 79.0052 + \text{Wt}\%(\text{LiC}_{12}) / 151.0694) \quad (1)$$

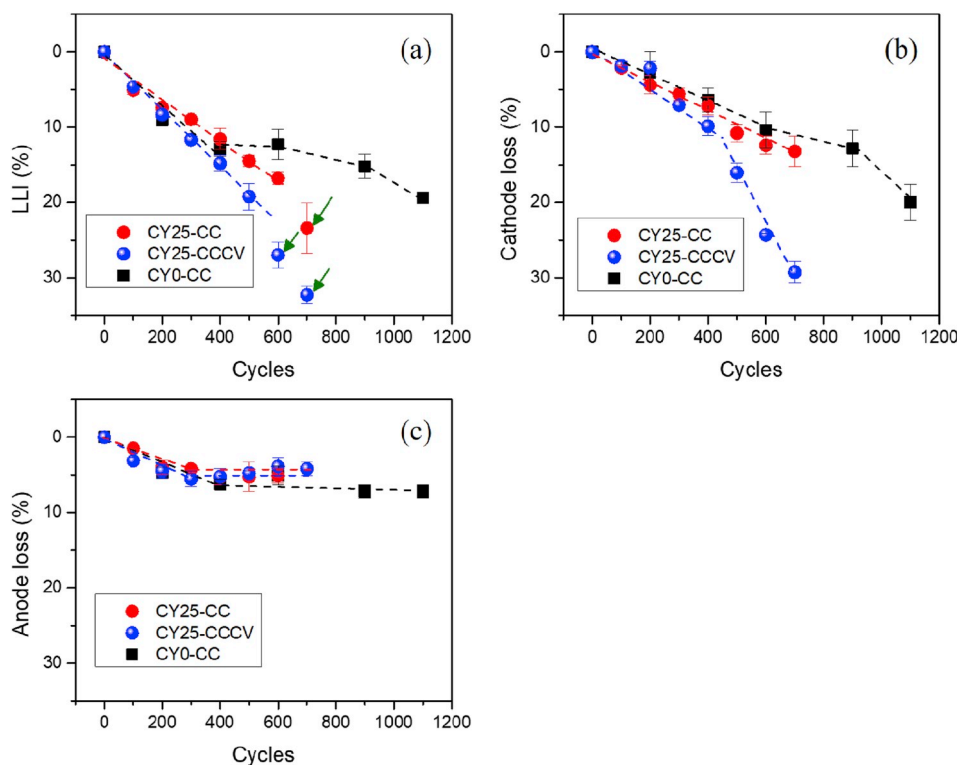


Fig. 3. Cell degradation factors using the dV/dQ method, dashed lines are the trend lines. Green arrows indicate the abrupt decrease of the LLI. Error bars indicate standard deviations between two independent cell measurements. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

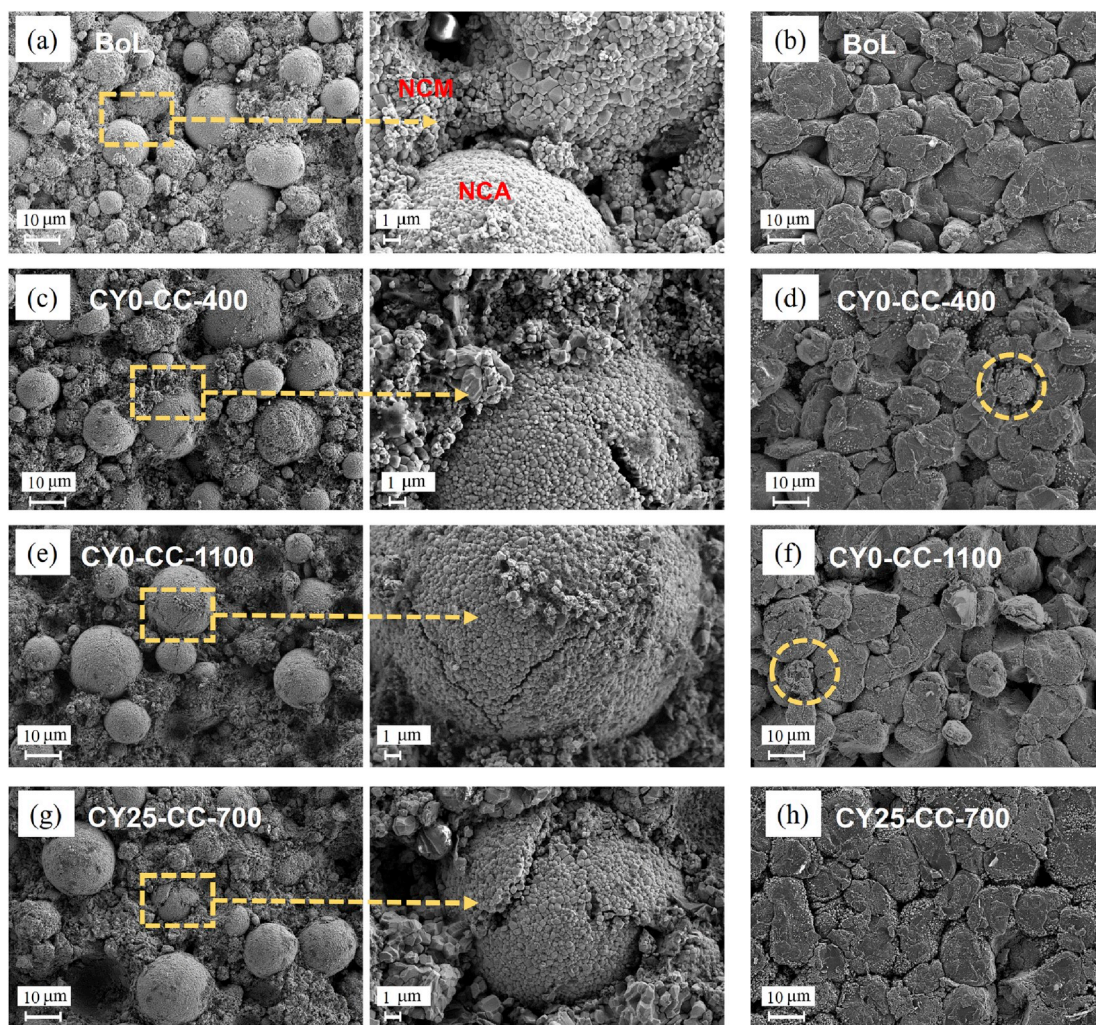


Fig. 4. Comparison of SEM morphology of the electrodes from BoL and fatigued cells.

$$\text{mol}(\text{LiC}_{12}) = (\text{Wt.}(\text{LiC}_{12}) / 151.0694) / (\text{Wt.}(\text{LiC}_6) / 79.0052 + \text{Wt.}(\text{LiC}_{12}) / 151.0694) \quad (2)$$

$$x \text{ in } \text{Li}_x\text{C}_6 = (\text{mol}(\text{LiC}_{12}) + \text{mol}(\text{LiC}_6)) / (2 * \text{mol}(\text{LiC}_{12}) + \text{mol}(\text{LiC}_6)) \quad (3)$$

3.5. Degradation mechanisms by impedance analysis

As illustrated in Fig. 6a, a typical impedance curve for the used cell can be divided into three parts along the frequency range: the first semi-circle at high frequencies, a depressed arc in the medium range, and a slope diffusion tail at low frequencies. An equivalent circuit model (ECM) is utilized for the interpretation and fitting of the experimental impedance spectra. R_0 is the real part of the impedance at zero crossing, which is the ohmic resistance. R_1/CPE_1 in parallel denotes the migration of lithium ions through the solid electrolyte interphase in the first high frequency range. The semi-circle in the medium frequency range is accounted for the charge transfer process, and modeled in ECM as charge transfer resistance R_2 in parallel with a double-layer capacitance CPE_2 . The low frequency slope is associated with Warburg impedance (W) corresponding to semi-infinite diffusion. It is described in Fig. 6, where W_{eff} is the Warburg coefficient and f is the excitation frequency.

In our cell degradation studies, the impedances of the BoL and fatigued cells are measured at the same temperature of 25 °C. The series of

impedance spectra at various fatigued stages are shown in Fig. S5. To compare the impedance of all fatigued cells, the parameters of the proposed ECM are identified with the 50% SoC data, and the results are plotted in Fig. 6b ~ 6e. From Fig. 6b, for all cells cycled at 25 °C an obvious increase of ohmic resistance R_0 is observed after 400 cycles (marked by the green arrows). As reported in Ref. [5], the increase of R_0 is also attributed to the lack of electrolyte and increased separator resistance. R_1 is used to describe the SEI resistance. Because the formation of SEI film between the anode and the electrolyte consumes the lithium and electrolyte decomposition products, the R_1 growth contributes the reduction of the cyclable lithium in the cell. There is a sudden R_1 increase as shown by the green arrow in Fig. 6c, which is probably caused by the growth of the SEI layer. The R_2 covers the faradaic charge transfer reactions at the electrolyte/electrode interface, it is apparent that the charge transfer resistance R_2 grows faster than R_0 and R_1 in Fig. 6d, which means a faster deterioration of the ionic kinetics in the cell. The solid state diffusion process of lithium species in the subsequent intercalation is represented by Warburg impedance W . An increasing trend of W_{eff} is displayed in Fig. 6e, indicating that the hindrance of lithium diffusion also increases with the cycle numbers. The R_2 and W are supposed to show overlapping degradation mechanisms: first, loss of active electrode area by degradation processes lead to the increase of both R_2 and W [39], then second, the lower numbers of transferable lithium will decrease the dynamics during the intercalation and de-intercalation processes for the fatigued cells. The R_0 and R_1 are more “sensitive” to the cell capacities sudden drop, e.g. there is almost

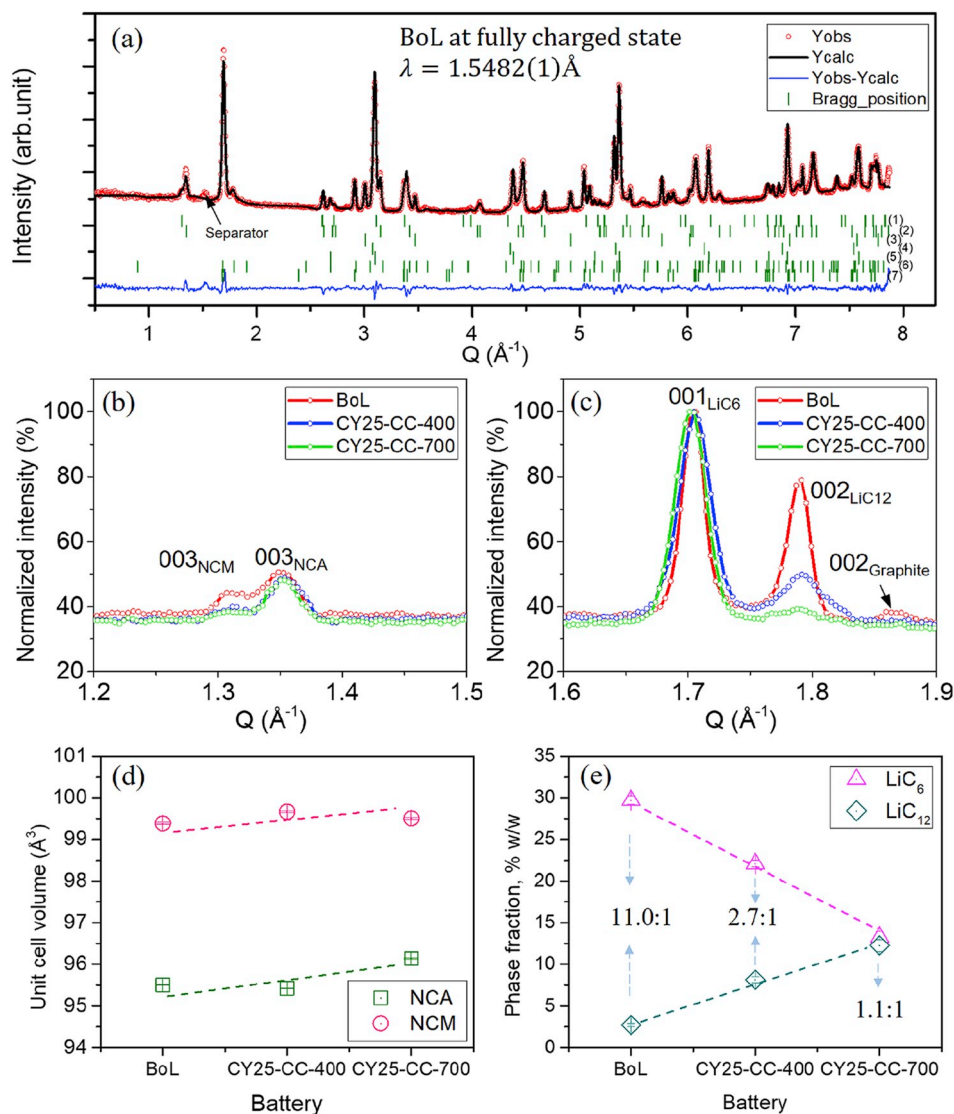


Fig. 5. *In situ* neutron powder diffraction results of the BoL and selected fatigued cells at fully charged state. (a) Neutron powder diffraction pattern for the BoL cell including the Rietveld refinement. (b) Selected range of neutron powder diffraction data focusing on the Bragg peak NCM (003) and NCA (003) phases. (c) Selected range for LiC₆ and LiC₁₂. (d) Calculated unit cell volumes. (e) Calculated phase weight ratio of LiC₆ and LiC₁₂.

Table 3

Evolution of crystal structural parameters of the NCA, NCM cathode from neutron powder diffraction patterns for the 18650-type commercial battery at fully charged state (4.2 V).

Cell	NCA phase (R $\bar{3}$ m)			NCM phase (R $\bar{3}$ m)		
	Lattice parameter <i>a</i> (Å)	Lattice parameter <i>c</i> (Å)	Unit cell volume (Å ³)	Lattice parameter <i>a</i> (Å)	Lattice parameter <i>c</i> (Å)	Unit cell volume (Å ³)
BoL	2.8097 (1)	13.9691 (12)	95.501 (8)	2.8197 (2)	14.4339 (38)	99.383 (28)
CY25-CC-400	2.8096 (1)	13.9589 (10)	95.426 (7)	2.8184 (2)	14.4884 (22)	99.664 (17)
CY25-CC-700	2.8124 (1)	14.0348(21)	96.139 (15)	2.8164 (2)	14.4858 (25)	99.506 (19)

no R_0 and R_1 increase when cells are in the linear degradation part, the sudden increase of R_0 and R_1 occur with the capacity sudden degradation as marked by the green arrows for CY25-CC and CY25-CC-CV in Fig. 6b and c. The R_2 and W_{eff} show good correlation with the capacity fade for all cells in the course of cycling.

To distinguish the cathode resistance and the anode resistance from the full cell impedance spectrum, the impedance spectra of the cathode and anode from the opened cells are measured using the half-cells. Fig. S6 shows the impedance spectra at BoL and in fatigued states (CY25-CC, 700 cycles) for cathode and anode. The impedance spectra of

the electrodes are measured at 25 °C and approximately 50% SoC. It shows a significant growth of the medium frequency arc with increasing cycle numbers in Fig. S6a, indicating that the charge transfer resistance dominates the total impedance increase of the cathode. The hindrance of electrochemical reaction rises along with reduced acceptance of lithium species and loss of active surface area due to the lithiated cathode material loss. A slight impedance arc marked by the green arrow appears as shown in the enlarged plot in Fig. S6a, which might express the new cathode surface films. The anode impedance arc does not show obvious change in Fig. S6b. The first semi-circle as marked by the green arrow

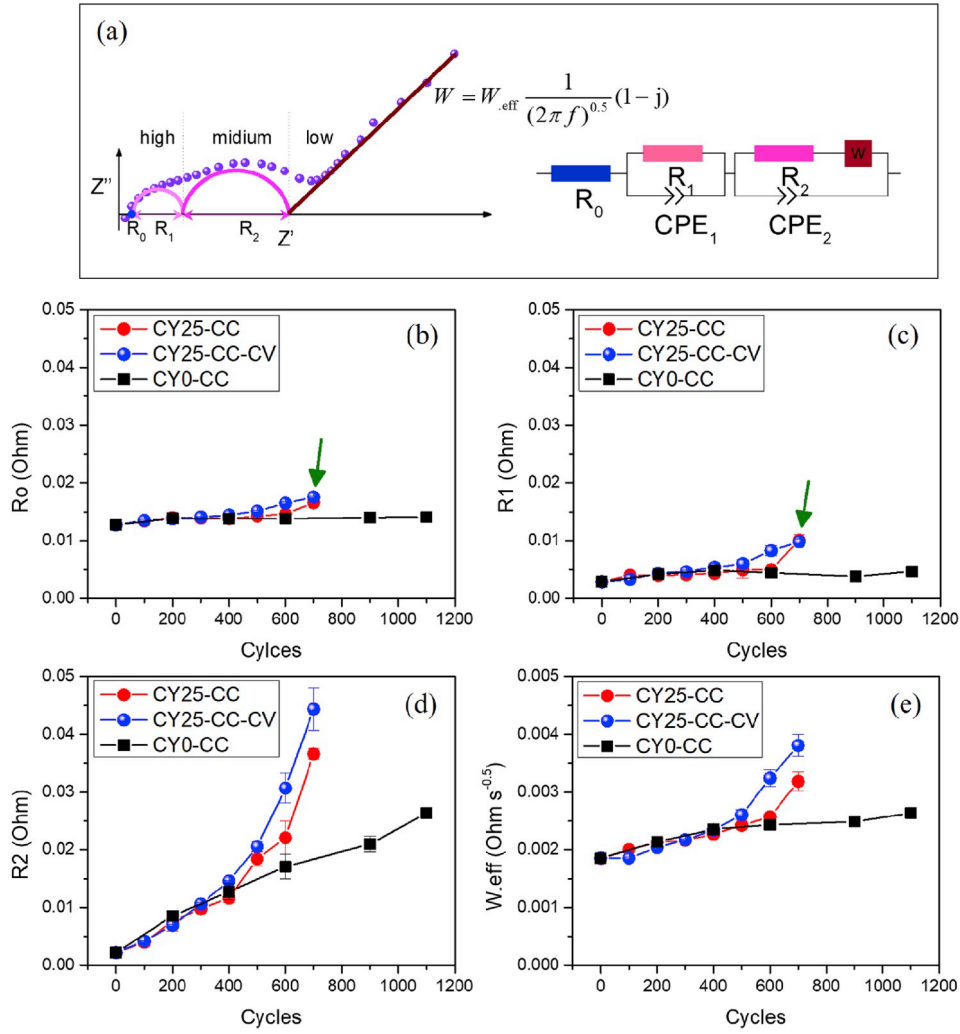


Fig. 6. The impedance results of the cycled cells. (a) Typical Nyquist impedance spectrum of the used battery. Identified ECM parameters, i.e. R_0 (a), R_1 (b), R_2 (c), and W_{eff} (d), from the impedance spectroscopy of the BoL and fatigued cells.

shows an increasing trend with cycle numbers, which might refer to the SEI film growth at the surface of the graphite particles. Accordingly, it can be concluded that the increase of the full cell mid-frequency arc upon cycling is predominantly ascribed to the cathode electrode degradation.

3.6. Correlation investigation for the degradation factors

The correlation coefficient r_{xy} is employed in this study to clarify the statistical connections of battery degradation factors for the fatigued cells (CY25-CC, CY25-CC-CV and CY0-CC). It is calculated by:

$$r_{xy} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \sqrt{\sum (Y_i - \bar{Y})^2}} \quad (4)$$

The calculated values are in the range from -1 to $+1$, where -1 indicates the strongest negative agreement and $+1$ represents the strongest positive agreement. The variables X_i and Y_i in equation (4) are the degradation factors obtained from dV/dQ and AC impedance methods, i is the cycle number, $\bar{X}$ and $\bar{Y}$ are the averaged values of the variables. All the correlation values are summarized in Table 4.

As expected, the cathode loss has strong correlation with the LLI, indicating that the loss of lithiated lithium sites (inactive phase) contributes a lot to the LLI. Furthermore, the cathode loss is correlated positively with R_2 growth ($r_{xy} = 0.91$), and it is consistent with the coin

cell impedance results of Section 3.5 in which the cathode impedance dominates the charge transfer impedance arc in the full cell. The anode loss does not present a significant correlation with any impedance parameters from the correlation analysis shown in Table 4, which can be interpreted by the non-linear anode degradation as depicted in Fig. 3. Fig. 7 is plotted to illustrate the correlations of the degradation factors with $r_{xy} > 0.85$.

From Fig. 7a, the LLI shows a linear relationship with the active cathode loss, and LLI is faster than the active cathode loss, indicating that not only the lithiated cathode loss but also some extra lithium loss, e.g. SEI formation, cathode surface film formation, contributes to the LLI. The dependence of R_0 on the cathode loss is displayed in Fig. 7b. There is not obvious R_0 change in the range of 0%–10% of the cathode loss for the three cycling protocols. Strong relationships between cathode loss and R_2 ($r_{xy} = 0.91$) as well as W_{eff} ($r_{xy} = 0.92$) are plotted in Fig. 7c and d. It means that the electrochemical reaction and diffusion processes are slowed down over cycling due to the active cathode loss. It can be supported by the SEM morphology in Section 3.3. The micro-cracks in the cathode particles indeed serve as paths for electrolyte infiltration and promote the buildup of the surface reconstruction layer that hinders lithium migration. As marked by the green arrows in Fig. 7c and d, there are several points deviating from the trend lines, indicating that some other degradation factors (side reactions) might participate to these parts. All the impedance parameters against the LLI are illustrated in Fig. 7e ~ 7h. The dependence of R_0 on the LLI in Fig. 7e is similar with

Table 4Correlation r_{xy} matrix of the degradation factors.

Active cathode loss	0.25	0.94	0.86	0.75	0.91	0.92
Active anode loss		0.34	0.08	0.01	0.27	0.19
LLI			0.91	0.86	0.96	0.97
R_0				0.92	0.92	0.94
				R_1	0.87	0.91
					R_2	0.98
						W_{eff}

its dependence on the cathode loss (Fig. 7b). When the LLI reaches around 20% for CY25-CC and CY25-CC-CV, the R_0 shows an increased trend with LLI. The same variation can also be seen for the R_1 in Fig. 7f. It can be explained by the new SEI formation consuming the mobile lithium and the electrolyte. The buildup of the SEI is reflected by the R_1 increase, and the increase of R_0 is attributed to lack of electrolyte. The charge transfer resistance R_2 and W_{eff} show an increasing trend with the LLI as depicted in Fig. 7g and h. According to the electrochemical theory, the R_2 is related to the exchange current density and it can be described by Eq. (5) after the derivation from the Butler-Volmer equation [41,60]:

$$R_2 = \frac{RT}{nFi_0} \quad (5)$$

where R is gas constant, T is the absolute temperature in Kelvin, n is the number of exchanged electrons in the underlying electrochemical reaction per active ion, F is the Faraday constant, i_0 is the exchange current

density. From Newman's porous electrode theory [61], the i_0 depends on the lithium ion concentration in the electrodes and electrolyte, the i_0 and SoC are calculated from the following equations:

$$i_0 = Fk_0 \sqrt{c_l(c_{s,max} - c_{s,surf})c_{s,surf}} \quad (6)$$

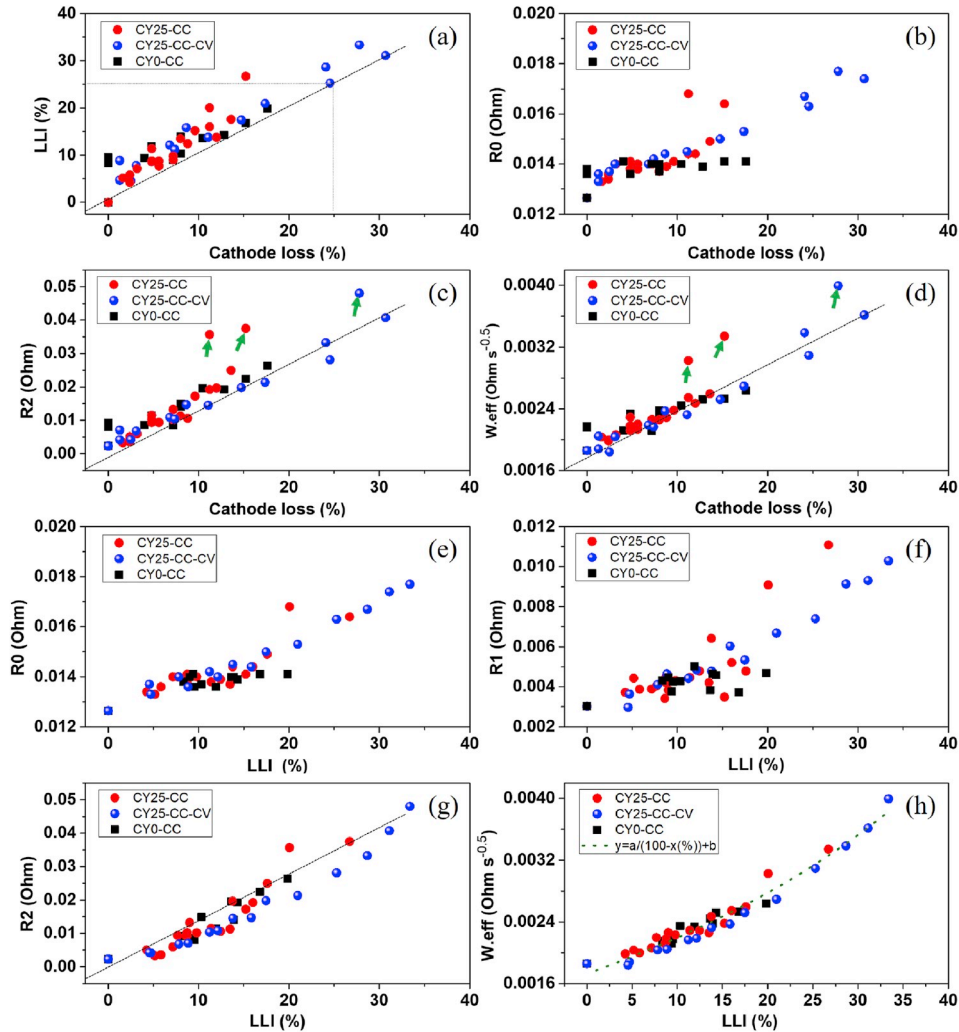
$$SoC = \frac{c_{s,surf}}{c_{s,max}} \quad (7)$$

where k_0 is the electrochemical reaction rate constant, c_l is the lithium ion concentration in the electrolyte, $c_{s,surf}$ is the concentration of lithium ions on the solid phase surface, $c_{s,max}$ is the maximum concentration of lithium ions in the solid phase.

According to equation (5) ~ (7), the charge transfer resistance can be given as:

$$R_2 = \frac{1}{c_{s,max}} \frac{1}{\sqrt{c_l}} \frac{RT}{nF^2k_0 \sqrt{(1 - SoC)SoC}} \quad (8)$$

Equation (8) reveals that at defined temperature and SoC, the charge transfer resistance has an inverse relationship with $c_{s,max}$ and $\sqrt{c_l}$. The $c_{s,max}$ is affected by the acceptance of lithium ions and active surface area in the electrode. The exposure of particle interior to electrolyte leads to the accelerated cathode loss and eventually undermine the mechanical integrity of the cathode particles. Either the loss of lithiated lithium sites or the microcracks in the cathode is supposed to decrease the $c_{s,max}$. Moreover, the amount of mobile lithium in the electrolyte is corrected

**Fig. 7.** Correlation between thermodynamic degradation variables and kinetic variables.

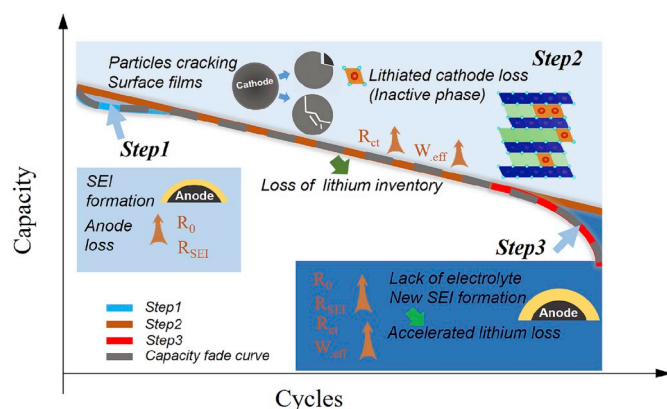


Fig. 8. Schematic of the battery degradation mechanisms during battery capacity fade.

by the total amount of lithium. The LLI leads to a reduction of c_i , and consequentially enlarges R_2 .

The Warburg resistance reflects the diffusion of lithium within the crystalline structure of the electrodes, degradation of the structure of the active material, e.g. particle segregation, phase segregation, will hinder the intercalation of ionic species. As elucidated in Fig. 7d, the W_{eff} is correlated with the cathode loss. Nonetheless, there is a sudden increase of W_{eff} in CY25-CC and CY25-CC-CV as displayed by the green arrows in Fig. 7d, meaning the W_{eff} is not only dependent on the cathode loss but also dependent on other factors. It can be seen that the W_{eff} has a strong mathematic relationship with LLI for all cells as shown in Fig. 7h. The diffusion process is usually assumed to obey Fick's second law, which is commonly used to describe the diffusion dynamics inside the electrode [29]. It is assumed that electrodes consist of multiple spherical particles, and the diffusion is driven by the lithium ion concentration difference between the inner parts of the particles and surface-near regions. A common expression of W_{eff} is given as [62,63]:

$$W_{eff} = \frac{RT}{\sqrt{2}n^2F^2\sqrt{D_s c_s}} \quad (9)$$

where D_s is the solid diffusion coefficient. c_s is the average lithium ion concentration in solid particles, which is equal to $c_{s,surf}$ when the electrode is in electrochemical equilibrium state. Obviously, the diffusion coefficient reduces with the degradation of the active material structure caused by the particle cracking. c_s is proportional to the total amount of mobile lithium at a defined SoC. Conclusively, the cathode loss and LLI in the cell will decrease the driving force for diffusion, which leads to the growth of W_{eff} in Fig. 7d and h. By comparing the results in Fig. 7d and h, the dependence of W_{eff} on the LLI is more notable than that on the cathode loss, especially in the region where the LLI exceeds 20%. Thus, an empirical equation, $y = a/(100-x(\%))+b$, in which a and b are the fitting parameters, is tentatively to describe the functional correlation between W_{eff} and LLI with high coefficient of determination (0.96) as shown by the dot curve in Fig. 7h. The strong numerical relationship between W_{eff} and LLI is independent of the cycling conditions, which could provide a promising countermeasure for the complicated battery diagnosis and prognosis.

3.7. Summary of degradation mechanisms during capacity fade

For the cycling protocols used in this investigation, three fatigue steps are identified during the cell cycling as shown in Fig. 8. The capacity fade curve is separated by the three steps schematically. A fast capacity loss for CY0-CC, CY25-CC, and CY25-CC-CV can be observed at the beginning of the cycling, which mainly results from some initial SEI formation and anode loss consuming the inventory of mobile lithium. An increase of SEI resistance and ohmic resistance are observed in step 1.

The main reasons for the capacity loss of all cycled cells are the active cathode loss and LLI in the course of cycling as shown in step 2. They are growing almost simultaneously with increasing cycles. The active cathode loss includes the lithiated material loss and the particle cracking. The lithiated cathode loss means that the lithium is trapped in the cathode under the controlled battery cycling voltage window, which leads to some part of cathode becoming inactive. The microcracks in the cathode secondary particles cause the disintegration between the cathode particles and even results in complete contact loss with accompanied inactivity of some parts of active material [52–55]. The formation of microcracks also creates new surface films, which might consume the active lithium and hinder the lithium migration, consequently, contributing to continuous rise of the charge transfer and diffusion impedance. In the end, the lithium loss into the cathode (inactive phase) and formation of surface films after particle cracking are deemed to be the reasons behind the LLI in step 2. In step 3, an accelerated capacity drop for CY25-CC and CY25-CC-CV occurs at a residual capacity of 80%. The CY0-CC has not reached to step 3 because its capacity loss is 15% on 1100 cycles. In addition to the on-going fatigue mechanisms in step 2, an accelerated active lithium loss could contribute to the capacity fast drop in step 3, as obvious increases of ohmic resistance and solid electrolyte interphase resistance are observed when the battery capacity fade falls beneath 20%.

4. Conclusion

The batteries are cycled using three different protocols. Three thermodynamic degradation factors, LLI, active cathode loss, and anode loss, are calculated from the dV/dQ . It is found that the LLI and active cathode loss play the dominating roles for battery degradation. *In situ* neutron powder diffraction and *post-mortem* analysis are performed to enhance the analysis of the battery fatigue mechanisms. Expansion of the unit cell volume of the cathode is in good agreement with the lithiated cathode loss (inactive phase). The LLI is also verified by the calculation of lithium content x in Li_xC_6 from the relative phase ratio of LiC_6 and LiC_{12} in the fatigued battery at fully charged state. The cathode particles of the fatigued cells present microcracks from SEM images, which leads to the active cathode loss and LLI at the same time because of the new surface film formation. The surface modification hinders the lithium insertion/extraction processes, consequently, causes the impedance increase. Impedance parameters are revealed from the fitting of impedance spectra. The gap between battery fatigue mechanism factors and impedance parameters is bridged using correlation analysis. (1) The correlation between active cathode loss and LLI follows a linear trend over cycle number, proving that the loss of lithiated lithium sites in the cathode primarily contributes to the LLI. (2) When the capacity fade goes beneath 20%, the accelerated LLI in the cell is ascribed to the new SEI formation, which is evidenced by the increase of the SEI resistance and ohmic resistance. (3) The increase of the charge transfer resistance is highly dependent on both active cathode loss and LLI. (4) A strong mathematic relationship between LLI and W_{eff} is revealed with high creditability independent of the used cycling conditions. It presents a potential value in the field of on-board battery SoH estimation, while, it can provide a fast diagnostic and prediction method for the battery second-life application.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2019.227575>.

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